Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001
B.Sc Degree Examination – APRIL 2021

MT616 – COMPLEX ANALYSIS – II

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. State fundamental theorem of algebra.
2. Write Taylor series.
3. Find the isolated singular points of the function $\frac{z+1}{z^3(z^2+1)}$.
4. Determine the number of roots of the equation $z^7 - 4z + z - 1 = 0$.
5. Define linear fractional transformation.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. State and prove Liouvilles' theorem.
7. Suppose that $z_n = x_n + iy_n$ (n = 1, 2, …..) and z = x+iy then prove that $\lim_{n\to\infty} z_n = z$ if and only if $\lim_{n\to\infty} x_n = x$ and $\lim_{n\to\infty} y_n = y$.
8. State and prove Cauchy's residue theorem.
9. State and prove Jordan Lemma.
10. Find the linear fractional transformation that maps the points $z_1 = 2$, $z_2 = i$ and $z_3 = -2$ on the points $w_1 = 1$, $w_2 = i$, $w_3 = -1$.

SECTION – C (5X10=50)

Answer **ALL** Questions

11. a) State and prove Cauchy integral formula.

(or)

b) Suppose that f(z) is analytic throughout a neighborhood $|z - z_0| < \in$ of a point z_0. If $|f(z)| \leq |f(z_0)|$ for each point z in that neighborhood then show that f(z) has the constant value $f(z_0)$ throughout the neighborhood.

12. a) State and prove Laurent series.

(or)

b) If a series $\sum_{n=0}^{\infty} a_n (z - z_0)^n$ converges to f(z) at all points interior to some circle $|z - z_0| = R$ then prove that it is the Taylor Series expansion for f in powers of $z - z_0$.

13. a) Show that $\int_C \frac{dz}{z(z-2)^4} = \frac{-\pi i}{8}$ where C is the positively oriented circle $|z - 2| = 1$.

(or)

b) Find the residue at z= 0 of the following functions.

i) $\frac{1}{z+z^2}$ ii) $z \cos\left(\frac{1}{z}\right)$ iii) $\frac{z-\sin z}{z}$

14. a) Show that $\int_0^{2\pi} \frac{d\theta}{1+a\sin\theta} = \frac{2\pi}{\sqrt{1-a^2}}$ $(-1 < a < 1)$.

(or)

b) State and prove argument principle.

15. a) Determine all linear fractional transformation that map the upper half plane Im z >0 on to the open disk $|w| < 1$ and the boundary Im z = 0 on the boundary $|w| = 1$.

(or)

b) Prove that the mapping $w = \frac{1}{z}$ transforms circles and lines in to circles and lines.

****